

URI. For example, URI parameters could be used to launch the app to a user's account page, but we recommend that you never use them to directly modify the user's account.

ⓘ Note

If you are creating a new URI scheme name for your app, be sure to follow the guidance in [RFC 4395](#). This ensures that your name meets the standards for URI schemes.

ⓘ Note

When launched via Protocol Contract, make sure that Back button takes the user back to the screen that launched the app and not to the app's previous content.

We recommend that apps create a new XAML **Frame** for each activation event that opens a new Uri target. This way, the navigation backstack for the new XAML **Frame** will not contain any previous content that the app might have on the current window when suspended.

If you decide that you want your apps to use a single XAML **Frame** for Launch and Protocol Contracts, clear the pages on the **Frame** navigation journal before navigating to a new page. When launched via Protocol Contract, consider including UI into your apps that allows the user to go back to the top of the app.

Related topics

Complete sample app

- [Association launching sample](#)

Concepts

- [Default Programs](#)
- [File Type and URI Associations Model](#)

Tasks

- [Launch the default app for a URI](#)
- [Handle file activation](#)

Guidelines

- [Guidelines for file types and URIs](#)

Reference

- [AppX Package Manifest](#)
- [Windows.ApplicationModel.Activation.ProtocolActivatedEventArgs](#)
- [Windows.UI.Xaml.Application.OnActivated](#)

Launch an app for results

Article • 05/26/2023

Important APIs

- [LaunchUriForResultsAsync](#)
- [ValueSet](#)

Learn how to launch an app from another app and exchange data between the two. This is called *launching an app for results*. The example here shows you how to use [LaunchUriForResultsAsync](#) to launch an app for results.

New app-to-app communication APIs in Windows 10 make it possible for Windows apps (and Windows Web apps) to launch an app and exchange data and files. This enables you to build mash-up solutions from multiple apps. Using these new APIs, complex tasks that would have required the user to use multiple apps can now be handled seamlessly. For example, your app could launch a social networking app to choose a contact, or launch a checkout app to complete a payment process.

The app that you'll launch for results will be referred to as the launched app. The app that launches the app will be referred to as the calling app. For this example you will write both the calling app and the launched app.

Step 1: Register the protocol to be handled in the app that you'll launch for results

In the Package.appxmanifest file of the launched app, add a protocol extension to the `<Application>` section. The example here uses a fictional protocol named `test-app2app`.

The `ReturnResults` attribute in the protocol extension accepts one of these values:

- **optional**—The app can be launched for results by using the [LaunchUriForResultsAsync](#) method, or not for results by using [LaunchUriAsync](#). When you use **optional**, the launched app must determine whether it was launched for results. It can do that by checking the `OnActivated` event argument. If the argument's `IActivatedEventArgs.Kind` property returns `ActivationKind.ProtocolForResults`, or if the type of the event argument is `ProtocolActivatedEventArgs`, the app was launched via `LaunchUriForResultsAsync`.
- **always**—The app can be launched only for results; that is, it can respond only to [LaunchUriForResultsAsync](#).

- **none**—The app cannot be launched for results; it can respond only to [LaunchUriAsync](#).

In this protocol-extension example, the app can be launched only for results. This simplifies the logic inside the **OnActivated** method, discussed below, because we have to handle only the "launched for results" case and not the other ways that the app could be activated.

XML

```
<Applications>
  <Application ...>

    <Extensions>
      <uap:Extension Category="windows.protocol">
        <uap:Protocol Name="test-app2app" ReturnResults="always">
          <uap:DisplayName>Test app-2-app</uap:DisplayName>
        </uap:Protocol>
      </uap:Extension>
    </Extensions>

  </Application>
</Applications>
```

Step 2: Override Application.OnActivated in the app that you'll launch for results

If this method does not already exist in the launched app, create it within the `App` class defined in `App.xaml.cs`.

In an app that lets you pick your friends in a social network, this function could be where you open the people-picker page. In this next example, a page named **LaunchedForResultsPage** is displayed when the app is activated for results. Ensure that the **using** statement is included at the top of the file.

C#

```
using Windows.ApplicationModel.Activation;
...
protected override void OnActivated(IActivatedEventArgs args)
{
    // Window management
    Frame rootFrame = Window.Current.Content as Frame;
    if (rootFrame == null)
    {
        rootFrame = new Frame();
        Window.Current.Content = rootFrame;
    }
}
```

```

    }

    // Code specific to launch for results
    var protocolForResultsArgs = (ProtocolForResultsActivatedEventArgs)args;
    // Open the page that we created to handle activation for results.
    rootFrame.Navigate(typeof(LaunchedForResultsPage),
        protocolForResultsArgs);

    // Ensure the current window is active.
    Window.Current.Activate();
}

```

Because the protocol extension in the Package.appxmanifest file specifies **ReturnResults** as **always**, the code just shown can cast `args` directly to

ProtocolForResultsActivatedEventArgs with confidence that only **ProtocolForResultsActivatedEventArgs** will be sent to **OnActivated** for this app. If your app can be activated in ways other than launching for results, you can check whether **IActivatedEventArgs.Kind** property returns **ActivationKind.ProtocolForResults** to tell whether the app was launched for results.

Step 3: Add a ProtocolForResultsOperation field to the app you launch for results

C#

```
private Windows.System.ProtocolForResultsOperation _operation = null;
```

You'll use the **ProtocolForResultsOperation** field to signal when the launched app is ready to return the result to the calling app. In this example, the field is added to the **LaunchedForResultsPage** class because you'll complete the launch-for-results operation from that page and will need access to it.

Step 4: Override OnNavigatedTo() in the app you launch for results

Override the **OnNavigatedTo** method on the page that you'll display when your app is launched for results. If this method does not already exist, create it within the class for the page defined in `<pagename>.xaml.cs`. Ensure that the following **using** statement is included at the top of the file:

C#

```
using Windows.ApplicationModel.Activation
```

The **NavigationEventArgs** object in the **OnNavigatedTo** method contains the data passed from the calling app. The data may not exceed 100KB and is stored in a **ValueSet** object.

In this example code, the launched app expects the data sent from the calling app to be in a **ValueSet** under a key named **TestData**, because that's what the example's calling app is coded to send.

C#

```
using Windows.ApplicationModel.Activation;
...
protected override void OnNavigatedTo(NavigationEventArgs e)
{
    var protocolForResultsArgs = e.Parameter as
ProtocolForResultsActivatedEventArgs;
    // Set the ProtocolForResultsOperation field.
    _operation = protocolForResultsArgs.ProtocolForResultsOperation;

    if (protocolForResultsArgs.Data.ContainsKey("TestData"))
    {
        string dataFromCaller = protocolForResultsArgs.Data["TestData"] as
string;
    }
}
...
private Windows.System.ProtocolForResultsOperation _operation = null;
```

Step 5: Write code to return data to the calling app

In the launched app, use **ProtocolForResultsOperation** to return data to the calling app. In this example code, a **ValueSet** object is created that contains the value to return to the calling app. The **ProtocolForResultsOperation** field is then used to send the value to the calling app.

C#

```
ValueSet result = new ValueSet();
result["ReturnedData"] = "The returned result";
_operation.ReportCompleted(result);
```

Step 6: Write code to launch the app for results and get the returned data

Launch the app from within an async method in your calling app as shown in this example code. Note the **using** statements, which are necessary for the code to compile:

```
C#

using System.Threading.Tasks;
using Windows.System;
...

async Task<string> LaunchAppForResults()
{
    var testAppUri = new Uri("test-app2app:"); // The protocol handled by
    the launched app
    var options = new LauncherOptions();
    options.TargetApplicationPackageFamilyName = "67d987e1-e842-4229-9f7c-
    98cf13b5da45_yd7nk54bq29ra";

    var inputData = new ValueSet();
    inputData["TestData"] = "Test data";

    string theResult = "";
    LaunchUriResult result = await
    Windows.System.Launcher.LaunchUriForResultsAsync(testAppUri, options,
    inputData);
    if (result.Status == LaunchUriStatus.Success &&
        result.Result != null &&
        result.Result.ContainsKey("ReturnedData"))
    {
        ValueSet theValues = result.Result;
        theResult = theValues["ReturnedData"] as string;
    }
    return theResult;
}
```

In this example, a **ValueSet** containing the key **TestData** is passed to the launched app. The launched app creates a **ValueSet** with a key named **ReturnedData** that contains the result returned to the caller.

You must build and deploy the app that you'll launch for results before running your calling app. Otherwise, **LaunchUriResult.Status** will report **LaunchUriStatus.AppUnavailable**.

You'll need the family name of the launched app when you set the **TargetApplicationPackageFamilyName**. One way to get the family name is to make the following call from within the launched app:

C#

```
string familyName = Windows.ApplicationModel.Package.Current.Id.FamilyName;
```

Remarks

The example in this how-to provides a "hello world" introduction to launching an app for results. The key things to note are that the new [LaunchUriForResultsAsync](#) API lets you asynchronously launch an app and communicate via the [ValueSet](#) class. Passing data via a **ValueSet** is limited to 100KB. If you need to pass larger amounts of data, you can share files by using the [SharedStorageAccessManager](#) class to create file tokens that you can pass between apps. For example, given a **ValueSet** named `inputData`, you could store the token to a file that you want to share with the launched app:

C#

```
inputData["ImageFileToken"] = SharedStorageAccessManager.AddFile(myFile);
```

Then pass it to the launched app via [LaunchUriForResultsAsync](#).

Related topics

- [LaunchUri](#)
- [LaunchUriForResultsAsync](#)
- [ValueSet](#)

Choose and save tones using the ms-tonepicker URI scheme

Article • 10/20/2022

This topic describes how to use the **ms-tonepicker**: URI scheme. This URI scheme can be used to:

- Determine if the tone picker is available on the device.
- Display the tone picker to list available ringtones, system sounds, text tones, and alarm sounds; and get a tone token which represents the sound the user selected.
- Display the tone saver, which takes a sound file token as input and saves it to the device. Saved tones are then available via the tone picker. Users can also give the tone a friendly name.
- Convert a tone token to its friendly name.

ms-tonepicker: URI scheme reference

This URI scheme does not pass arguments via the URI scheme string, but instead passes arguments via a [ValueSet](#). All strings are case-sensitive.

The sections below indicate which arguments should be passed to accomplish the specified task.

Task: Determine if the tone picker is available on the device

C#

```
var status = await Launcher.QueryUriSupportAsync(new Uri("ms-tonepicker:"),
    LaunchQuerySupportType.UriForResults,
    "Microsoft.Tonepicker_8wekyb3d8bbwe");

if (status != LaunchQuerySupportStatus.Available)
{
    // the tone picker is not available
}
```

Task: Display the tone picker

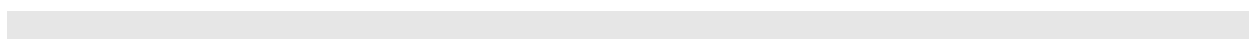
The arguments you can pass to display the tone picker are as follows:

Parameter	Type	Required	Possible values	Description
Action	string	yes	"PickRingtone"	Opens the tone picker.
CurrentToneFilePath	string	no	An existing tone token.	The tone to show as the current tone in the tone picker. If this value is not set, the first tone on the list is selected by default. This is not, strictly speaking, a file path. You can get a suitable value for <code>CurrenttoneFilePath</code> from the <code>ToneToken</code> value returned from the tone picker.
TypeFilter	string	no	"Ringtones", "Notifications", "Alarms", "None"	Selects which tones to add to the picker. If no filter is specified then all tones are displayed.

The values that are returned in [LaunchUriResults.Result](#):

Return values	Type	Possible values	Description
Result	Int32	0-success. 1-cancelled. 7-invalid parameters. 8 - no tones match the filter criteria. 255 - specified action is not implemented.	The result of the picker operation.
ToneToken	string	The selected tone's token. The string is empty if the user selects default in the picker.	This token can be used in a toast notification payload, or can be assigned as a contact's ringtone or text tone. The parameter is returned in the ValueSet only if Result is 0.
DisplayName	string	The specified tone's friendly name.	A string that can be shown to the user to represent the selected tone. The parameter is returned in the ValueSet only if Result is 0.

Example: Open the tone picker so that the user can select a tone



C#

```
LauncherOptions options = new LauncherOptions();
options.TargetApplicationPackageFamilyName =
    "Microsoft.Tonepicker_8wekyb3d8bbwe";

ValueSet inputData = new ValueSet() {
    { "Action", "PickRingtone" },
    { "TypeFilter", "Ringtones" } // Show only ringtones
};

LaunchUriResult result = await Launcher.LaunchUriForResultsAsync(new
    Uri("ms-tonepicker:"), options, inputData);

if (result.Status == LaunchUriStatus.Success)
{
    Int32 resultCode = (Int32)result.Result["Result"];
    if (resultCode == 0)
    {
        string token = result.Result["ToneToken"] as string;
        string name = result.Result["DisplayName"] as string;
    }
    else
    {
        // handle failure
    }
}
```

Task: Display the tone saver

The arguments you can pass to display the tone saver are as follows:

Parameter	Type	Required	Possible values	Description
Action	string	yes	"SaveRingtone"	Opens the picker to save a ringtone.
ToneFileSharingToken	string	yes	SharedStorageAccessManager file sharing token for the ringtone file to save.	Saves a specific sound file as a ringtone. The supported content types for the file are mpeg audio and x-ms-wma audio.
DisplayName	string	no	The specified tone's friendly name.	Sets the display name to use when saving the specified ringtone.

The values that are returned in [LaunchUriResults.Result](#):

Return values	Type	Possible values	Description
Result	Int32	0-success. 1-cancelled by user. 2-Invalid file. 3-Invalid file content type. 4-file exceeds maximum ringtone size (1MB in Windows 10). 5-File exceeds 40 second length limit. 6-File is protected by digital rights management. 7-invalid parameters.	The result of the picker operation.

Example: Save a local music file as a ringtone

C#

```
LauncherOptions options = new LauncherOptions();
options.TargetApplicationPackageFamilyName =
    "Microsoft.Tonepicker_8wekyb3d8bbwe";

ValueSet inputData = new ValueSet() {
    { "Action", "SaveRingtone" },
    { "ToneFileSharingToken",
        SharedStorageAccessManager.AddFile(myLocalFile) }
};

LaunchUriResult result = await Launcher.LaunchUriForResultsAsync(new
    Uri("ms-tonepicker:"), options, inputData);

if (result.Status == LaunchUriStatus.Success)
{
    Int32 resultCode = (Int32)result.Result["Result"];

    if (resultCode == 0)
    {
        // no issues
    }
    else
    {
        switch (resultCode)
        {
            case 2:
                // The specified file was invalid
                break;
            case 3:
                // The specified file's content type is invalid
                break;
            case 4:
```

```

        // The specified file was too big
        break;
    case 5:
        // The specified file was too long
        break;
    case 6:
        // The file was protected by DRM
        break;
    case 7:
        // The specified parameter was incorrect
        break;
    }
}
}

```

Task: Convert a tone token to its friendly name

The arguments you can pass to get the friendly name of a tone are as follows:

Parameter	Type	Required	Possible values	Description
Action	string	yes	"GetToneName"	Indicates that you want to get the friendly name of a tone.
ToneToken	string	yes	The tone token	The tone token from which to obtain a display name.

The values that are returned in [LaunchUriResults.Result](#):

Return value	Type	Possible values	Description
Result	Int32	0-The picker operation succeeded. 7-Incorrect parameter (for example, no ToneToken provided). 9-Error reading the name for the specified token. 10-Unable to find specified tone token.	The result of the picker operation.
DisplayName	string	The tone's friendly name.	Returns the selected tone's display name. This parameter is only returned in the ValueSet if Result is 0.

Example: Retrieve a tone token from `Contact.RingToneToken` and display its friendly name in the contact card.

C#

```
using (var connection = new AppServiceConnection())
{
    connection.AppServiceName = "ms-tonepicker-nameprovider";
    connection.PackageFamilyName = "Microsoft.Tonepicker_8wekyb3d8bbwe";
    AppServiceConnectionStatus connectionStatus = await
connection.OpenAsync();
    if (connectionStatus == AppServiceConnectionStatus.Success)
    {
        var message = new ValueSet() {
            { "Action", "GetToneName" },
            { "ToneToken", token)
        };
        AppServiceResponse response = await
connection.SendMessageAsync(message);
        if (response.Status == AppServiceResponseStatus.Success)
        {
            Int32 resultCode = (Int32)response.Message["Result"];
            if (resultCode == 0)
            {
                string name = response.Message["DisplayName"] as string;
            }
            else
            {
                // handle failure
            }
        }
        else
        {
            // handle failure
        }
    }
}
```

Launch the Windows Settings app

Article • 07/24/2024

Important APIs

- [LaunchUriAsync](#)
- [PreferredApplicationPackageFamilyName](#)
- [DesiredRemainingView](#)

Learn how to launch the Windows Settings app. This topic describes the `ms-settings:` URI scheme. Use this URI scheme to launch the Windows Settings app to specific settings pages.

Launching to the Settings app is an important part of writing a privacy-aware app. If your app can't access a sensitive resource, we recommend providing the user a convenient link to the privacy settings for that resource. For more info, see [Guidelines for privacy-aware apps](#).

How to launch the Settings app

To launch the **Settings** app, use the `ms-settings:` URI scheme as shown in the following examples.

XAML hyperlink control

In this example, a Hyperlink XAML control is used to launch the privacy settings page for the microphone using the `ms-settings:privacy-microphone` URI.

XML

```
<!--Set Visibility to Visible when access to the microphone is denied -->
<TextBlock x:Name="LocationDisabledMessage" FontStyle="Italic"
           Visibility="Collapsed" Margin="0,15,0,0"
           TextWrapping="Wrap" >
    <Run Text="This app is not able to access the microphone. Go to "
/>
    <Hyperlink NavigateUri="ms-settings:privacy-microphone">
        <Run Text="Settings" />
    </Hyperlink>
    <Run Text=" to check the microphone privacy settings."/>
</TextBlock>
```

Calling LaunchUriAsync

Alternatively, your app can call the [LaunchUriAsync](#) method to launch the **Settings** app. This example shows how to launch to the privacy settings page for the camera using the `ms-settings:privacy-webcam` URI.

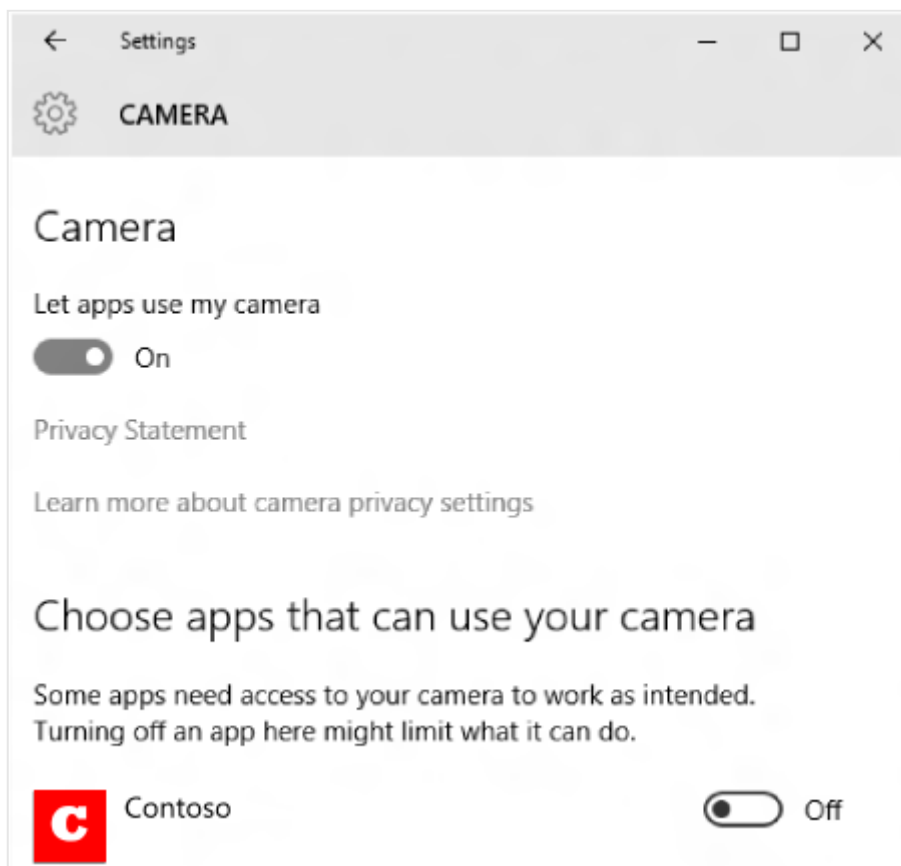
CS

```
bool result = await Windows.System.Launcher.LaunchUriAsync(new Uri("ms-  
settings:privacy-webcam"));
```

C++/WinRT

```
bool result = co_await  
Windows::System::Launcher::LaunchUriAsync(Windows::Foundation::Uri(L"ms-  
settings:privacy-webcam"));
```

The code above launches the privacy settings page for the camera:



For more info about launching URIs, see [Launch the default app for a URI](#).

ms-settings: URI scheme reference

The following sections describe different categories of ms-settings URIs used to open various pages of the Settings app:

- [Accounts](#)
- [Apps](#)
- [Control Center](#)
- [Cortana](#)
- [Devices](#)
- [Ease of access](#)
- [Extras](#)
- [Family Group](#)
- [Gaming](#)
- [Mixed reality](#)
- [Network and internet](#)
- [Personalization](#)
- [Phone](#)
- [Privacy](#)
- [Search](#)
- [Sound](#)
- [Surface Hub](#)
- [System](#)
- [Time and language](#)
- [Update and security](#)
- [User accounts](#)

Note

The availability of some settings pages varies by Windows version and SKU. The notes column also captures additional requirements that must be met for a page to be available.

Accounts

 Expand table

Settings page	URI
Access work or school	ms-settings:workplace
Email & app accounts	ms-settings:emailandaccounts
Family & other people	ms-settings:otherusers

Settings page	URI
Set up a kiosk	ms-settings:assignedaccess
Sign-in options	ms-settings:signinoptions ms-settings:signinoptions-dynamiclock
Sync your settings	ms-settings:sync ms-settings:backup (Backup page deprecated in Windows 11)
Windows Hello setup	ms-settings:signinoptions-launchfaceenrollment ms-settings:signinoptions-launchfingerprintenrollment
Your info	ms-settings:yourinfo

Apps

 Expand table

Settings page	URI
Apps & Features	ms-settings:appsfeatures
App features	ms-settings:appsfeatures-app (Reset, manage add-on & downloadable content, etc. for the app) To access this page with a URI, use the ms-settings:appsfeatures-app URI and pass an optional parameter of the <i>package family name</i> of the app.
Apps for websites	ms-settings:appsforwebsites
Default apps	ms-settings:defaultapps (Behavior introduced in Windows 11, version 21H2 (with 2023-04 Cumulative Update) or 22H2 (with 2023-04 Cumulative Update), or later.) Append the query string parameter in the following formats using the Uri-escaped name of an app to directly launch the default settings page for that app: - registeredAppMachine= <Uri-escaped per machine installed name of app> - registeredAppUser= <Uri-escaped per user installed name of app> - registeredAUMID= <Uri-escaped Application User Model ID> For more information, see Launch the Default Apps settings page.
Default browser settings	ms-settings:defaultbrowsersettings (Deprecated in Windows 11)

Settings page	URI
Manage optional features	ms-settings:optionalfeatures
Offline Maps	ms-settings:maps ms-settings:maps-downloadmaps (Download maps)
Startup apps	ms-settings:startupapps
Video playback	ms-settings:videoplayback

Control Center

 Expand table

Settings page	URI
Control center	ms-settings:controlcenter

Cortana

 Expand table

Settings page	URI
Cortana across my devices	ms-settings:cortana-notifications
More details	ms-settings:cortana-moredetails
Permissions & History	ms-settings:cortana-permissions
Searching Windows	ms-settings:cortana-windowssearch
Talk to Cortana	ms-settings:cortana-language ms-settings:cortana ms-settings:cortana-talktocortana

Important

Cortana voice assistance in Windows as a standalone app was retired in the spring of 2023. For more information, see [End of support for Cortana](#).

ⓘ Note

This Settings section on desktop will be called Search when the PC is set to regions where Cortana is not currently available or Cortana has been disabled. Cortana-specific pages (Cortana across my devices, and Talk to Cortana) will not be listed in this case.

Devices

 Expand table

Settings page	URI
AutoPlay	ms-settings:autoplay
Bluetooth	ms-settings:bluetooth
Connected Devices	ms-settings:connecteddevices
Default camera	ms-settings:camera (Behavior deprecated in Windows 10, version 1809 and later)
Camera settings	ms-settings:camera (Behavior introduced in Windows 11, build 22000 and later) Append the query string parameter <code>cameraId</code> set to the Uri-escaped symbolic link name of a camera device to directly launch the settings for that camera. For more information, see Launch the camera settings page .
Mouse & touchpad	ms-settings:mousetouchpad (touchpad settings only available on devices that have a touchpad)
Pen & Windows Ink	ms-settings:pen
Printers & scanners	ms-settings:printers
Touch	ms-settings:devices-touch
Touchpad	ms-settings:devices-touchpad (only available if touchpad hardware is present)
Text Suggestions	ms-settings:devicestyping-hwkbtextsuggestions
Typing	ms-settings:typing
USB	ms-settings:usb

Settings page	URI
Wheel	ms-settings:wheel (only available if a Surface Dial device is paired)
Your phone	ms-settings:mobile-devices

Ease of access

 Expand table

Settings page	URI
Audio	ms-settings:easeofaccess-audio
Closed captions	ms-settings:easeofaccess-closedcaptioning
Color filters	ms-settings:easeofaccess-colorfilter ms-settings:easeofaccess-colorfilter-adaptivecolorlink ms-settings:easeofaccess-colorfilter-bluelightlink
Display	ms-settings:easeofaccess-display
Eye control	ms-settings:easeofaccess-eyecontrol
Fonts	ms-settings:fonts
High contrast	ms-settings:easeofaccess-highcontrast
Keyboard	ms-settings:easeofaccess-keyboard
Magnifier	ms-settings:easeofaccess-magnifier
Mouse	ms-settings:easeofaccess-mouse
Mouse pointer & touch	ms-settings:easeofaccess-mousepointer
Narrator	ms-settings:easeofaccess-narrator ms-settings:easeofaccess-narrator-isautostartenabled
Speech	ms-settings:easeofaccess-speechrecognition
Text cursor	ms-settings:easeofaccess-cursor
Visual Effects	ms-settings:easeofaccess-visualeffects

Extras

[Expand table](#)

Settings page	URI
Extras	ms-settings:extras (only available if "settings apps" have been installed, for example, by a 3rd party)

Family Group

[Expand table](#)

Settings page	URI
Family Group	ms-settings:family-group

Gaming

[Expand table](#)

Settings page	URI
Game bar	ms-settings:gaming-gamebar
Game DVR	ms-settings:gaming-gamedvr
Game Mode	ms-settings:gaming-gamemode
Playing a game full screen	ms-settings:quietmomentsgame
TruePlay	ms-settings:gaming-trueplay (As of Windows 10, version 1809 (10.0; Build 17763), this feature is removed from Windows)

Mixed reality

ⓘ Note

These settings are only available if the [Mixed Reality Portal](#) [↗](#) app is installed.

[Expand table](#)

Settings page	URI
Audio and speech	ms-settings:holographic-audio
Environment	ms-settings:privacy-holographic-environment
Headset display	ms-settings:holographic-headset
Uninstall	ms-settings:holographic-management
Startup and desktop	ms-settings:holographic-startupanddesktop

Network and internet

[Expand table](#)

Settings page	URI
Network & internet	ms-settings:network-status
Advanced settings	ms-settings:network-advancedsettings
Airplane mode	ms-settings:network-airplanemode ms-settings:proximity
Cellular & SIM	ms-settings:network-cellular
Dial-up	ms-settings:network-dialup
DirectAccess	ms-settings:network-directaccess (only available if DirectAccess is enabled)
Ethernet	ms-settings:network-ethernet
Manage known networks	ms-settings:network-wifisettings
Mobile hotspot	ms-settings:network-mobilehotspot
Proxy	ms-settings:network-proxy
VPN	ms-settings:network-vpn
Wi-Fi	ms-settings:network-wifi (only available if the device has a wifi adapter)
Wi-Fi provisioning	ms-settings:wifi-provisioning

Personalization

[Expand table](#)

Settings page	URI
Background	ms-settings:personalization-background
Choose which folders appear on Start	ms-settings:personalization-start-places
Colors	ms-settings:personalization-colors ms-settings:colors
Customize Copilot key on keyboard	ms-settings:personalization-textinput-copilot-hardwarekey
Dynamic Lighting	ms-settings:personalization-lighting
Glance	ms-settings:personalization-glance (Deprecated in Windows 10, version 1809 and later)
Lock screen	ms-settings:lockscreen
Navigation bar	ms-settings:personalization-navbar (Deprecated in Windows 10, version 1809 and later)
Personalization (category)	ms-settings:personalization
Start	ms-settings:personalization-start
Taskbar	ms-settings:taskbar
Text input	ms-settings:personalization-textinput
Touch Keyboard	ms-settings:personalization-touchkeyboard
Themes	ms-settings:themes

Phone

[Expand table](#)

Settings page	URI
Your phone	ms-settings:mobile-devices ms-settings:mobile-devices-addphone ms-settings:mobile-devices-addphone-direct (Opens Your Phone app)
Device Usage	ms-settings:deviceusage

Privacy

 Expand table

Settings page	URI
Accessory apps	ms-settings:privacy-accessoryapps (Deprecated in Windows 10, version 1809 and later)
Account info	ms-settings:privacy-accountinfo
Activity history	ms-settings:privacy-activityhistory
Advertising ID	ms-settings:privacy-advertisingid (Deprecated in Windows 10, version 1809 and later)
App diagnostics	ms-settings:privacy-appdiagnostics
Automatic file downloads	ms-settings:privacy-automaticfiledownloads
Background Apps	ms-settings:privacy-backgroundapps (Deprecated in Windows 11, 21H2 and later) Note: In Windows 11, the background app permissions are accessed individually. To view the permissions, go to Apps->Installed apps and then select "..." on a modern app and choose Advanced options. The advanced page is present for modern apps, and the Background apps permissions section will be present unless a group policy has been set or the user's global toggle value (the deprecated setting from Windows 10) is set. To access this page with a URI, use the <code>ms-settings:appsfeatures-app</code> URI and pass an optional parameter of the <i>package family name</i> of the app.
Background Spatial Perception	ms-settings:privacy-backgroundspatialperception
Calendar	ms-settings:privacy-calendar
Call history	ms-settings:privacy-callhistory
Camera	ms-settings:privacy-webcam
Contacts	ms-settings:privacy-contacts
Documents	ms-settings:privacy-documents

Settings page	URI
Downloads folder	ms-settings:privacy-downloadsfolder
Email	ms-settings:privacy-email
Eye tracker	ms-settings:privacy-eyetracker (requires eyetracker hardware)
Feedback & diagnostics	ms-settings:privacy-feedback
File system	ms-settings:privacy-broadfilesystemaccess
General	ms-settings:privacy or ms-settings:privacy-general
Graphics	ms-settings:privacy-graphicscaptureprogrammatic ms-settings:privacy-graphicscapturewithoutborder
Inking & typing	ms-settings:privacy-speechtyping
Location	ms-settings:privacy-location
Messaging	ms-settings:privacy-messaging
Microphone	ms-settings:privacy-microphone
Motion	ms-settings:privacy-motion
Music Library	ms-settings:privacy-musiclibrary
Notifications	ms-settings:privacy-notifications
Other devices	ms-settings:privacy-customdevices
Phone calls	ms-settings:privacy-phonecalls
Pictures	ms-settings:privacy-pictures
Radios	ms-settings:privacy-radios
Speech	ms-settings:privacy-speech
Tasks	ms-settings:privacy-tasks
Videos	ms-settings:privacy-videos
Voice activation	ms-settings:privacy-voiceactivation

Search

[Expand table](#)

Settings page	URI
Search	ms-settings:search
Search more details	ms-settings:search-moredetails
Search Permissions	ms-settings:search-permissions

Sound

[Expand table](#)

Settings page	URI
Volume mixer	ms-settings:apps-volume
Sound	ms-settings:sound
Sound devices	ms-settings:sound-devices
Default microphone	ms-settings:sound-defaultinputproperties
Default audio output	ms-settings:sound-defaultoutputproperties
Audio device properties (specific device)	ms-settings:sound-properties?endpointId={0.0.0.00000000}.{aaaaaaa-0000-1111-2222-bbbbbbbbbbbb} Note: User of URI must know the <code>endpointId</code> string to use.
Audio device properties (specific device)	ms-settings:sound-properties?interfaceId=\\?\SWD#MMDEVAPI#{3.0.0.00000003}.{bbbbbbbb-1111-2222-3333-cccccccccc}#{ccccccc-2222-3333-4444-ddddddddddd} Note: User of URI must know the <code>interfaceId</code> string to use and the string must be escaped correctly before sending.

Surface Hub

[Expand table](#)

Settings page	URI
Accounts	ms-settings:surfacehub-accounts
Session cleanup	ms-settings:surfacehub-sessioncleanup
Team Conferencing	ms-settings:surfacehub-calling
Team device management	ms-settings:surfacehub-devicemanagement
Welcome screen	ms-settings:surfacehub-welcome

System

[Expand table](#)

Settings page	URI
About	ms-settings:about
Advanced display settings	ms-settings:display-advanced (only available on devices that support advanced display options)
Battery Saver	ms-settings:batterysaver (only available on devices that have a battery, such as a tablet)
Battery Saver settings	ms-settings:batterysaver-settings (only available on devices that have a battery, such as a tablet)
Battery use	ms-settings:batterysaver-usedetails (only available on devices that have a battery, such as a tablet)
Clipboard	ms-settings:clipboard
Display	ms-settings:display
Default Save Locations	ms-settings:savelocations
Display	ms-settings:screenrotation
Duplicating my display	ms-settings:quietmomentspresentation
During these hours	ms-settings:quietmomentsscheduled
Encryption	ms-settings:deviceencryption
Energy recommendations	ms-settings:energyrecommendations (Available in February Moment update for Windows 11, Version 22H2, Build 22624 or later)

Settings page	URI
Focus assist	ms-settings:quiethours
Graphics Settings	ms-settings:display-advancedgraphics (only available on devices that support advanced graphics options)
Graphics Default Settings	ms-settings:display-advancedgraphics-default
Multitasking	ms-settings:multitasking ms-settings:multitasking-sgupdate
Night light settings	ms-settings:nightlight
Projecting to this PC	ms-settings:project
Shared experiences	ms-settings:crossdevice
Tablet mode	ms-settings:tabletmode (Removed in Windows 11)
Taskbar	ms-settings:taskbar
Notifications & actions	ms-settings:notifications
Remote Desktop	ms-settings:remotedesktop
Phone	ms-settings:phone (Deprecated in Windows 10, version 1809 and later)
Power & sleep	ms-settings:powersleep
Presence sensing	ms-settings:presence (Added in May Moment update for Windows 11, Version 22H2, Build 22624)
Storage	ms-settings:storage
Storage Sense	ms-settings:storagepolicies
Storage recommendations	ms-settings:storagerecommendations
Disks & volumes	ms-settings:disksandvolumes

Time and language

[Expand table](#)

Settings page	URI
Date & time	ms-settings:dateandtime

Settings page	URI
Japan IME settings	ms-settings:regionlanguage-jpnime (available if the Microsoft Japan input method editor is installed)
Region	ms-settings:regionformatting
Language	ms-settings:keyboard ms-settings:keyboard-advanced ms-settings:regionlanguage ms-settings:regionlanguage-bpmfime ms-settings:regionlanguage-cangjieime ms-settings:regionlanguage-chsime-wubi-udp ms-settings:regionlanguage-quickime ms-settings:regionlanguage-korime
Pinyin IME settings	ms-settings:regionlanguage-chsime-pinyin (available if the Microsoft Pinyin input method editor is installed) ms-settings:regionlanguage-chsime-pinyin-domainlexicon ms-settings:regionlanguage-chsime-pinyin-keyconfig ms-settings:regionlanguage-chsime-pinyin-udp
Speech	ms-settings:speech
Wubi IME settings	ms-settings:regionlanguage-chsime-wubi (available if the Microsoft Wubi input method editor is installed)
Add display language	ms-settings:regionlanguage-adddisplaylanguage
Language options	ms-settings:regionlanguage-languageoptions
Set display language	ms-settings:regionlanguage-setdisplaylanguage

Update and security

 Expand table

Settings page	URI
Activation	ms-settings:activation
Backup	ms-settings:backup (page removed in Windows 11; opens Sync)
Delivery Optimization	ms-settings:delivery-optimization ms-settings:delivery-optimization-activity

Settings page	URI
	ms-settings:delivery-optimization-advanced
Find My Device	ms-settings:findmydevice
For developers	ms-settings:developers
Recovery	ms-settings:recovery
Launch Security Key Enrollment	ms-settings:signinoptions-launchsecuritykeyenrollment
Troubleshoot	ms-settings:troubleshoot
Windows Security	ms-settings:windowsdefender
Windows Insider Program	ms-settings:windowsinsider (only present if user is enrolled in WIP) ms-settings:windowsinsider-optin
Windows Update	ms-settings:windowsupdate ms-settings:windowsupdate-action
Windows Update-Active hours	ms-settings:windowsupdate-activehours
Windows Update-Advanced options	ms-settings:windowsupdate-options
Windows Update-Optional updates	ms-settings:windowsupdate-optionalupdates
Windows Update-Restart options	ms-settings:windowsupdate-restartoptions
Windows Update-Seeker on demand	ms-settings:windowsupdate-seekerondemand
Windows Update-View update history	ms-settings:windowsupdate-history

User accounts

 Expand table

Settings page	URI
Provisioning	ms-settings:workplace-provisioning (only available if enterprise has deployed a provisioning package)
Repair token	ms-settings:workplace-repairtoken

Settings page	URI
Provisioning	ms-settings:provisioning (only available on mobile and if the enterprise has deployed a provisioning package)
Windows Anywhere	ms-settings:windowsanywhere (device must be Windows Anywhere-capable)

See also

[Launch the default app for a URI](#)

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Launch the Default Apps settings page

Article • 06/21/2024

Learn how to launch the Windows Settings app to display the Default Apps settings page from your app using the ms-settings URI scheme.

Windows defines a set of URIs that allow apps to launch the Windows Settings app and display a particular settings page. This article explains how to launch the Windows Settings app directly to the Default Apps settings page and, optionally, navigate directly to the settings for a specified default application. For more information, see [Launch the Windows Settings app](#).

The Default Apps settings URL

`ms-settings:defaultapps` launches the Windows Settings app and navigates to the Default Apps settings page. Starting with Windows 11, version 21H2 (with 2023-04 Cumulative Update), 22H2 (with 2023-04 Cumulative Update), and 23H2 or later, you can append an additional query string parameter in escaped URI format to launch directly to the settings page for a specific application.

There are three query string parameters. The query string parameter to be used depends on how the application was installed.

 Expand table

Query string parameter	Value to pass
registeredAppUser	Named value from HKEY_CURRENT_USER\Software\RegisteredApplications Use when the app was installed per user, and the registration for the app was written to HKEY_CURRENT_USER\Software\RegisteredApplications.
registeredAppMachine	Named value from HKEY_LOCAL_MACHINE\Software\RegisteredApplications Use when the app was installed per machine, and the registration for the app was written to HKEY_LOCAL_MACHINE\Software\RegisteredApplications.
registeredAUMID	Application User Model ID

Query string parameter	Value to pass
	Use when the app was registered with Package Manager using a manifest declaring that the app handles File Types (uap:FileTypeAssociation) or URI schemes (uap:Protocol).

ⓘ Note

To get the registeredAUMID query string parameter to work after an OS upgrade, an app may need to increment its TargetDeviceFamily...MaxVersionTested value in its manifest. This will ensure that the app is reindexed for the user, which in turn will update the appropriate definitions used to process the deep link via protocol activation. MaxVersionTested should be updated to `10.0.22000.1817` for Windows 11, version 21H2 or `10.0.22621.1555` for Windows 11, version 22H2.

In the following example, `LaunchUriAsync` is called to launch the Windows Settings app. The `ms-settings:defaultapps` Uri specifies that the Default Apps settings page should be shown. Next, the app that should be launched is determined. As an example, "Microsoft Edge" was registered by the app in `HKEY_LOCAL_MACHINE\Software\RegisteredApplications`. Since it is a per machine installed app, `registeredAppMachine` is the query string parameter that should be used. The optional query string parameter `registeredAppMachine` is set to the registered name, escaped with a call to `Uri.EscapeDataString`, to specify that the page for **Microsoft Edge** should be shown.

C#

```
private async void LaunchSettingsPage_Click(object sender, RoutedEventArgs e)
{
    bool result = await Windows.System.Launcher.LaunchUriAsync(new Uri("ms-
settings:defaultapps?registeredAppMachine=" +
Uri.EscapeDataString(("Microsoft Edge"))));
}
```

See also

[Launch the Windows Settings app](#)

[Launch the default app for a URI](#)

Feedback

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Using ms-windows-store URIs

Article • 07/31/2024

This topic describes the **ms-windows-store:** URI scheme. Your app can use this URI scheme to launch the Microsoft Store app to specific pages in the store by using the [LaunchUriAsync](#) method on Windows 10 and Windows 11.

For example, you can open the Store to the Games page using the following code:

C#

```
bool result = await Windows.System.Launcher.LaunchUriAsync(new Uri("ms-windows-store://navigatetopage/?Id=Gaming"));
```

Opening the Store to specific sections

You can launch the Store App to a specific page or section using the following options.

 Expand table

URI Scheme	Description	Notes
ms-windows-store://home	Launches the home page of the Store.	
ms-windows-store://navigatetopage/?Id=[vertical]	Launches a top-level vertical page of the store	Verticals include: <code>Home</code> , <code>Gaming</code> , <code>Entertainment</code> , <code>Productivity</code> , and <code>LOB</code> , but note that available verticals can vary by region.
ms-windows-store://downloadsandupdates	Launches the downloads and updates page.	Starting with the October 2021 update to the Store app, this will launch the Library page.
ms-windows-store://settings	Launches the Store settings page.	

Opening to a specific product

You can launch the Store directly to the product detail page (PDP) for a specific product by using the Product ID for an App. While the Store app on Windows 10 and Windows 11 still supports Package Family Name (PFN) and App IDs, these are deprecated and

may not be supported in the future. These values can be found in Partner Center on the App identity page in the Product management section for each app.

Starting with the October 2021 update to the Store app, there are two modes available for displaying the PDP. By default the Store app is opened to the product detail page. You can also launch the store into with a popup experience that displays a smaller PDP dialog with that only displays the essential details for your app and a single action button for users. For the popup experience, you can optionally specify the location of a window that the dialog should be centered above.

[Expand table](#)

URI Scheme	Description	Notes
ms-windows-store://pdp/?ProductId=9WZDNCRFHVJL	Launches the full product details page (PDP) for a product	This is the recommended way to link to a specific product.
ms-windows-store://pdp/?PFN=Microsoft.Office.OneNote_8wekyb3d8bbwe	Launches the full product details page (PDP) for a product	Using the package family name is deprecated.
ms-windows-store://pdp/?AppId=00001111-aaaa-2222-bbbb-3333cccc4444	Launches the full product details page (PDP) for a product	Using the App ID is deprecated.
ms-windows-store://pdp/?ProductId=9WZDNCRFHVJL&mode=mini	Launches the popup Store dialog experience	The popup experience only supports Product ID

Launching the rating and review experience for a product

To enable users to review your app, you can link to that PDP and launch directly into the rating and review dialog. Store ID is the recommended method to launch the Store app to a specific product detail page.

[Expand table](#)

URI Scheme	Description	Notes
ms-windows-store://review/?ProductId=9WZDNCRFHVJL	Launches the write a review experience for a product.	Using StoreId is recommended

URI Scheme	Description	Notes
ms-windows-store://review/?PFN=Microsoft.Office.OneNote_8wekyb3d8bbwe	Launches the write a review experience for a product.	Using product family name is deprecated.
ms-windows-store://review/?ApplId=11112222-bbbb-3333-cccc-4444dddd5555	Launches the write a review experience for a product.	Using productid is deprecated

Searching the Store

You can launch the Store app directly into search results with several supported methods for searching store content.

[Expand table](#)

URI Scheme	Description	Notes
ms-windows-store://assoc/?Tags=Photos_Rich_Media_Edit, Camera_Capture_App	Launches a search for products associated with one or more tags.	Tags must be separated by commas.
ms-windows-store://search/?query=OneNote	Launches a search for the specified query.	Spaces in the query are allowed.
ms-windows-store://browse/?type=Apps&cat=Health+%26+fitness	Launches a search for products in a category.	
ms-windows-store://publisher/?name=Microsoft Corporation	Launches a search for products from the specified publisher.	Spaces in the name are allowed.
ms-windows-store://assoc/?FileExt=pdf	Launches a search for products associated with a file extension.	
ms-windows-store://assoc/?Protocol=ms-word	Launches a search for products associated with a protocol.	

See also

[Launching, resuming, and background tasks](#)

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Launch the Windows Maps app

Article • 01/04/2023

Learn how to launch the Windows Maps app from your app. This topic describes the **bingmaps:**, **ms-drive-to:**, **ms-walk-to:**, and **ms-settings:** Uniform Resource Identifier (URI) schemes. Use these URI schemes to launch the Windows Maps app to specific maps, directions, and search results or to download Windows Maps offline maps from the Settings app.

Tip To learn more about launching the Windows Maps app from your app, download the [Universal Windows Platform \(UWP\) map sample](#) from the [Windows-universal-samples repo](#) on GitHub.

Introducing URIs

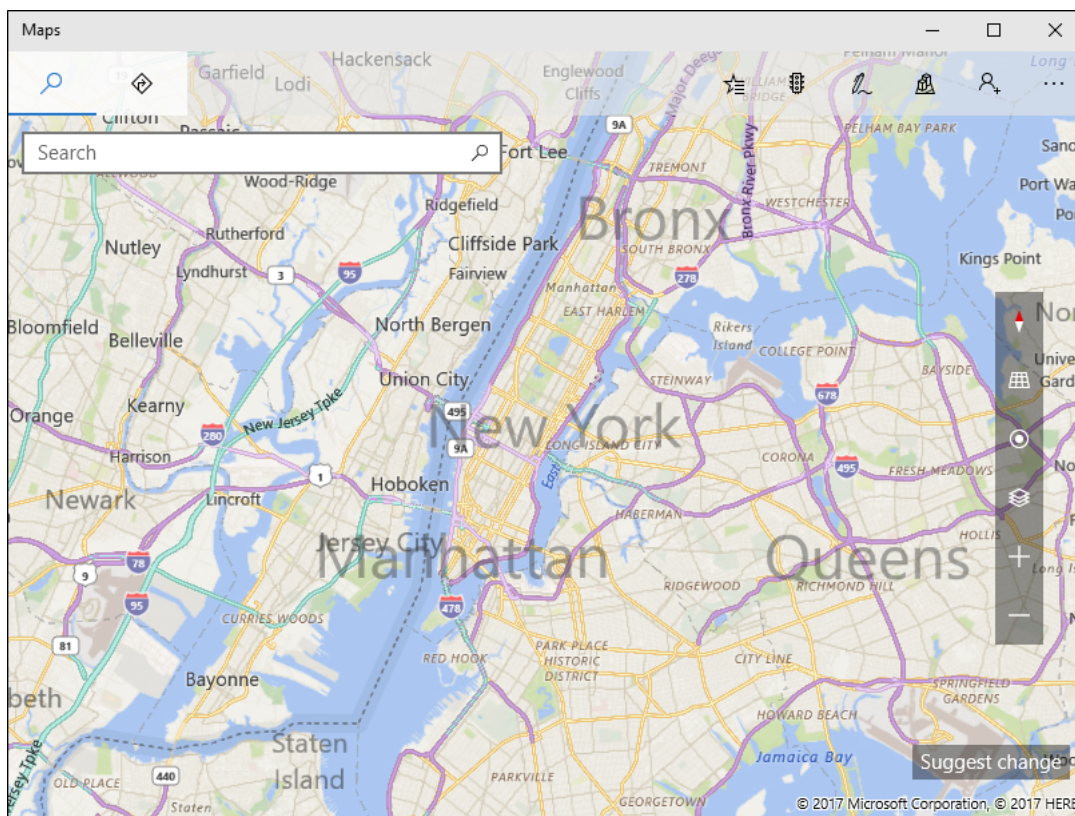
URI schemes let you open apps by clicking hyperlinks (or programmatically, in your app). Just as you can start a new email using **mailto:** or open a web browser using **http:**, you can open the Windows maps app using **bingmaps:**, **ms-drive-to:**, and **ms-walk-to:**.

- The **bingmaps:** URI provides maps for locations, search results, directions, and traffic.
- The **ms-drive-to:** URI provides turn-by-turn driving directions from your current location.
- The **ms-walk-to:** URI provides turn-by-turn walking directions from your current location.

For example, the following URI opens the Windows Maps app and displays a map centered over New York City.

XML

```
<bingmaps:?cp=40.726966~-74.006076>
```



Here is a description of the URI scheme:

bingmaps:?query

In this URI scheme, *query* is a series of parameter name/value pairs:

¶m1=value1¶m2=value2 ...

For a full list of the available parameters, see the [bingmaps:](#), [ms-drive-to:](#), and [ms-walk-to:](#) parameter reference. There are also examples later in this topic.

Launch a URI from your app

To launch the Windows Maps app from your app, call the [LaunchUriAsync](#) method with a **bingmaps**;, **ms-drive-to**;, or **ms-walk-to**: URI. The following example launches the same URI from the previous example. For more info about launching apps via URI, see [Launch the default app for a URI](#).

```
C#

// Center on New York City
var uriNewYork = new Uri(@"bingmaps:?cp=40.726966~-74.006076");

// Launch the Windows Maps app
var launcherOptions = new Windows.System.LauncherOptions();
launcherOptions.TargetApplicationPackageFamilyName = "Microsoft.WindowsMaps_8wekyb3d8bbwe";
var success = await Windows.System.Launcher.LaunchUriAsync(uriNewYork, launcherOptions);
```

In this example, the [LauncherOptions](#) class is used to help ensure the Windows Maps app is launched.

Display known locations

There are many options to control which part of the map to show. You can use the *cp* (center point) parameter with either the *rad* (radius) or the *lvl* (zoom level) parameters to show a location and choose how close to zoom in to it. When you use the *cp* parameter, you can also specify a *hdg* (heading) and *pit* (pitch) to control what direction to look. Another method is to use the *bb* (bounding box) parameter to provide the maximum south, east, north, and west coordinates of the area you want to show.

To control the type of view, use the *sty* (style) and *ss* (Streetside) parameters. The *sty* parameter lets you switch between road and aerial views. The *ss* parameter puts the map into a Streetside view. For more info about these and other parameters, see the [bingmaps: parameter reference](#).

Sample URI	Results
bingmaps:?	Opens the Maps app.
bingmaps:?cp=40.726966~-74.006076	Displays a map centered over New York City.
bingmaps:?cp=40.726966~-74.006076&lvl=10	Displays a map centered over New York City with a zoom level of 10.
bingmaps:?bb=39.719_-74.52~41.71_-73.5	Displays a map of New York City, which is the area specified in the bb argument.
bingmaps:?bb=39.719_-74.52~41.71_-73.5&cp=47~-122	Displays a map of New York City, which is the area specified in the bounding box argument. The center point for Seattle specified in the cp argument is ignored because <i>bb</i> is specified.
bingmaps:?collection=point.36.116584_-115.176753_Caesars%20Palace&lvl=16	Displays a map with a point named Caesars Palace (in Las Vegas) and sets the zoom level to 16.
bingmaps:?collection=point.40.726966_-74.006076_Some%255FBusiness	Displays a map with a point named Some_Business (in Las Vegas).
bingmaps:?cp=40.726966~-74.006076&trfc=1&sty=a	Displays a map of New York City with traffic on and aerial map style.
bingmaps:?cp=47.6204~-122.3491&sty=3d	Displays a 3D view of the Space Needle.
bingmaps:?cp=47.6204~-122.3491&sty=3d&rad=200&pit=75&hdg=165	Displays a 3D view of the Space Needle with a radius of 200m, a pitch of 75 degrees, and a heading of 165 degrees.
bingmaps:?cp=47.6204~-122.3491&ss=1	Displays a Streetside view of the Space Needle.

Display search results

When searching for places using the *q* parameter, we recommend making the terms as specific as possible and using the *cp*, *bb*, or *where* parameters to specify a search location. If you do not specify a search location and the user's current location isn't available, the search may not return meaningful results. Search results are displayed in the most appropriate map view. For more info about these and other parameters, see the [bingmaps: parameter reference](#).

Sample URI	Results
bingmaps:?q=1600%20Pennsylvania%20Ave,%20Washington,%20DC	Displays a map and searches for the address of the White House in Washington, D.C.
bingmaps:?q=coffee&where=Seattle	Searches for coffee in Seattle.
bingmaps:?cp=40.726966~-74.006076&where=New%20York	Searches for New York near the specified center point.

Sample URI	Results
bingmaps:?bb=39.719_-74.52~41.71_-73.5&q=pizza	Searches for pizza in the specified bounding box (that is, in New York City).

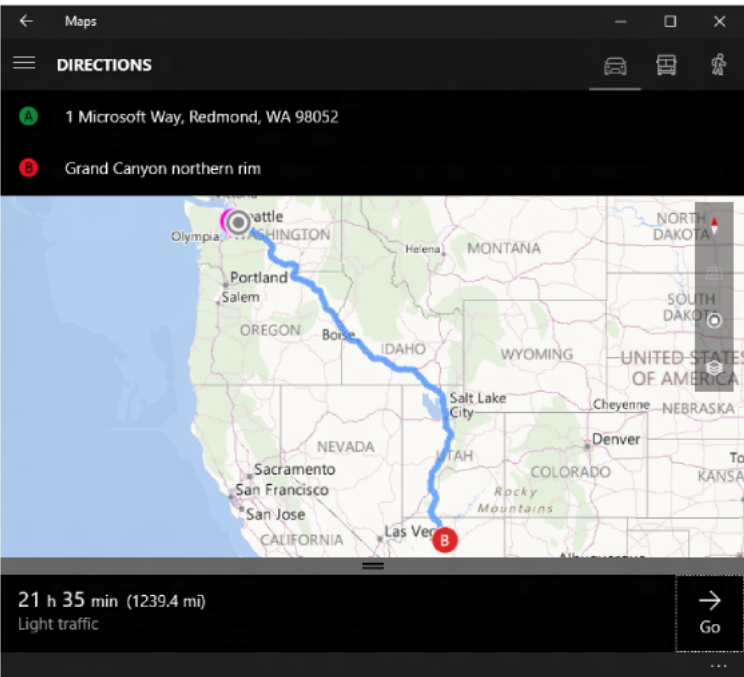
Display multiple points

Use the *collection* parameter to show a custom set of points on the map. If there is more than one point, a list of points is displayed. There can be up to 25 points in a collection and they are listed in the order provided. The collection takes precedence over search and directions requests. For more info about this parameter and others, see the [bingmaps: parameter reference](#).

Sample URI
bingmaps:?collection=point.36.116584_-115.176753_Caesars%20Palace
bingmaps:?collection=point.36.116584_-115.176753_Caesars%20Palace&lvl=16
bingmaps:?collection=point.36.116584_-115.176753_Caesars%20Palace~point.36.113126_-115.175188_The%20Bellagio&lvl=16&cp=36.114902~-115.176669
bingmaps:?collection=point.40.726966_-74.006076_Fake%255FBusiness%255Fwith%255FUnderscore
bingmaps:?collection=name.Hotel%20List~point.36.116584_-115.176753_Caesars%20Palace~point.36.113126_-115.175188_The%20Bellagio&lvl=16&cp=36.114902~-115.176669

Display directions and traffic

You can display directions between two points using the *rtp* parameter; those points can be either addresses or latitude and longitude coordinates. Use the *trfc* parameter to show traffic information. To specify the type of directions: driving, walking, or transit, use the *mode* parameter. If *mode* isn't specified, directions will be provided using the user's preferred mode of transportation. For more info about these parameters and others, see the [bingmaps: parameter reference](#).

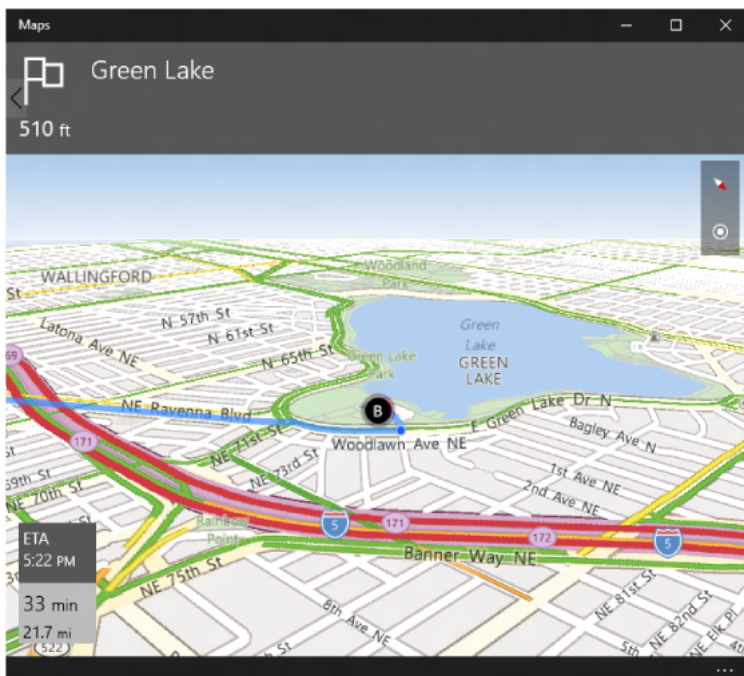


Sample URI	Results
bingmaps:?rtp=pos.44.9160_-110.4158~pos.45.0475_-109.4187	Displays a map with point-to-point directions. Because <i>mode</i> is not specified, directions will be provided using the user's mode of transportation preference.
bingmaps:?cp=43.0332~-87.9167&trfc=1	Displays a map centered over Milwaukee, WI with traffic.
bingmaps:?rtp=adr.One Microsoft Way, Redmond, WA 98052~pos.39.0731_-108.7238	Displays a map with directions from the specified address to the specified location.
bingmaps:?rtp=adr.1%20Microsoft%20Way,%20Redmond,%20WA,%2098052~pos.36.1223_-111.9495_Grand%20Canyon%20northern%20rim	Displays directions from 1 Microsoft Way, Redmond, WA, 98052 to the Grand Canyon's northern rim.
bingmaps:?rtp=adr.Davenport, CA~adr.Yosemite Village	Displays a map with driving directions from the specified location to the specified landmark.
bingmaps:?rtp=adr.Mountain%20View,%20CA~adr.San%20Francisco%20International%20Airport,%20CA&mode=d	Displays driving directions from Mountain View, CA to San Francisco International Airport, CA.
bingmaps:?rtp=adr.Mountain%20View,%20CA~adr.San%20Francisco%20International%20Airport,%20CA&mode=w	Displays walking directions from Mountain View, CA to San Francisco International Airport, CA.
bingmaps:?rtp=adr.Mountain%20View,%20CA~adr.San%20Francisco%20International%20Airport,%20CA&mode=t	Displays transit directions from Mountain View, CA to San Francisco International Airport, CA.

Display turn-by-turn directions

The **ms-drive-to:** and **ms-walk-to:** URI schemes let you launch directly into a turn-by-turn view of a route. These URI schemes can only provide directions from the user's current location. If you must provide directions between points that do not include the user's current location, use the **bingmaps:** URI scheme as described in the previous section. For more info about these URI schemes, see the [ms-drive-to:](#) and [ms-walk-to:](#) parameter reference.

Important When the **ms-drive-to:** or **ms-walk-to:** URI schemes are launched, the Maps app will check to see if the device has ever had a GPS location fix. If it has, then the Maps app will proceed to turn-by-turn directions. If it hasn't, the app will display the route overview, as described in [Display directions and traffic](#).



Sample URI	Results
<code>ms-drive-to:? destination.latitude=47.680504&destination.longitude=-122.328262&destination.name=Green Lake</code>	Displays a map with turn-by-turn driving directions to Green Lake from your current location.
<code>ms-walk-to:? destination.latitude=47.680504&destination.longitude=-122.328262&destination.name=Green Lake</code>	Displays a map with turn-by-turn walking directions to Green Lake from your current location.

Download offline maps

The **ms-settings:** URI scheme lets you launch directly into a particular page in the Settings app. While the **ms-settings:** URI scheme doesn't launch into the Maps app, it does allow you to launch directly to the Offline Maps page in the Settings app and displays a confirmation dialog to download the offline maps used by the Maps app. The URI scheme accepts a point specified by a latitude and longitude and automatically determines if there are offline maps available for a region containing that point. If the latitude and longitude passed happen to fall within multiple download regions, the confirmation dialog will let the user pick which of those regions to download. If offline maps are not available for a region containing that point, the offline Maps page in the Settings app is displayed with an error dialog.

Sample URI	Results
<code>ms-settings:maps-downloadmaps? latlong=47.6,-122.3</code>	Opens the Settings app to the Offline Maps page with a confirmation dialog displayed to download maps for the region containing the specified latitude-longitude point.

bingmaps: parameter reference

The syntax for each parameter in this table is shown by using Augmented Backus–Naur Form (ABNF).

Parameter	Definition	ABNF Definition and Example
-----------	------------	-----------------------------

Parameter	Definition	ABNF Definition and Example
cp	Center point	cp = "cp=" cpval cpval = degreeslat "~" degreeslon degreeslat = ["-"] 1*3DIGIT ["." 1*7DIGIT] degreeslon = ["-"] 1*2DIGIT ["." 1*7DIGIT] Example: cp=40.726966~-74.006076
bb	Bounding box	bb = "bb=" southlatitude "_" westlongitude "~" northlatitude "_" eastlongitude southlatitude = degreeslat northlatitude = degreeslat westlongitude = degreeslon eastlongitude = degreeslon degreeslat = ["-"] 13DIGIT ["." 17DIGIT] degreeslon = ["-"] 12DIGIT ["." 17DIGIT] Example: bb=39.719_-74.52~41.71_-73.5
where	Location	where = "where=" whereval whereval = 1*(ALPHA / DIGIT / "-" / "." / "_" / pct-encoded / "!" / "\$" / "'" / "(" / ")" / "*" / "+" / "," / ";" / ":" / "@" / "/" / "?") Example: where=1600%20Pennsylvania%20Ave,%20Washington,%20DC
q	Query Term	q = "q=" whereval Example: q=mexican%20restaurants
lvl	Zoom Level	lvl = "lvl=" 12DIGIT ["." 12DIGIT] Example: lvl=10.50

Parameter	Definition	ABNF Definition and Example
sty	Style	sty = "sty=" ("a" / "r" / "3d") Example: sty=a
rad	Radius	rad = "rad=" 1*8DIGIT Example: rad=1000
pit	Pitch	pit = "pit=" pitch Example: pit=60
hdg	Heading	hdg = "hdg=" heading Example: hdg=180

Parameter	Definition	ABNF Definition and Example
ss	Streetside	ss = "ss=" BIT Example: ss=1
trfc	Traffic	trfc = "trfc=" BIT Example: trfc=1
rtp	Route	rtp = "rtp=" (waypoint "~" [waypoint]) / ("~" waypoint) waypoint = ("pos." point) / ("adr." whereval) point = "point." pointval ["_" title] pointval = degreeslat "" degreeslon degreeslat = ["-"] 13DIGIT ["." 17DIGIT] degreeslon = ["-"] 12DIGIT ["." 17DIGIT] title = whereval whereval = 1(ALPHA / DIGIT / "-" / "." / "_" / pct-encoded / "!" / "\$" / "'" / "(" / ")" / "*" / "+" / "," / ";" / ":" / "@" / "/" / "?") Examples: rtp=adr.Mountain%20View,%20CA~adr.SFO rtp=adr.One%20Microsoft%20Way,%20Redmond,%20WA~pos.45.23423_-122.1232_My%20Picnic%20Spot

Parameter	Definition	ABNF Definition and Example

Parameter	Definition	ABNF Definition and Example
mode	Transportation mode	mode = "mode=" ("d" / "t" / "w") Example: mode=d

Parameter	Definition	ABNF Definition and Example
collection	Collection	<pre>collection = "collection="(name~"/")point["~"point] name = "name." whereval whereval = 1(ALPHA / DIGIT / "-" / ":" / "_" / pct-encoded / "!" / "\$" / "'" / "(" / ")" / "*" / "+" / "," / ";" / ":" / "@" / "/" / "?") point = "point." pointval ["_" title] pointval = degreeslat "" degreeslon degreeslat = ["-"] 13DIGIT ["." 17DIGIT] degreeslon = ["-"] 12DIGIT ["." 17DIGIT] title = whereval Example: collection=name.My%20Trip%20Stops~point.36.116584_-115.176753_Las%20Vegas~point.37.8268_-122.4798_Golden%20Gate%20Brid</pre>

ms-drive-to: parameter reference

The URI to launch a request for turn-by-turn driving directions does not need to be encoded and has the following format.

Note You don't specify the starting point in this URI scheme. The starting point is always assumed to be the current location. If you need to specify a starting point other than the current location, see [Display directions and traffic](#).

Parameter	Definition	Example	Details
destination.latitude	Destination latitude	Example: destination.latitude=47.6451413797194	The latitude of the destination. Valid latitude values are between -90 and +90 inclusive.
destination.longitude	Destination longitude	Example: destination.longitude=-122.141964733601	The longitude of the destination. Valid longitude values are between -180 and +180 inclusive.
destination.name	Name of the destination	Example: destination.name=Redmond, WA	The name of the destination. You do not have to encode the destination.name value.

ms-walk-to: parameter reference

The URI to launch a request for turn-by-turn walking directions does not need to be encoded and has the following format.

Note You don't specify the starting point in this URI scheme. The starting point is always assumed to be the current location. If you need to specify a starting point other than the current location, see [Display directions and traffic](#).

Parameter	Definition	Example	Details
destination.latitude	Destination latitude	Example: destination.latitude=47.6451413797194	The latitude of the destination. Valid latitude values are between -90 and +90 inclusive.
destination.longitude	Destination longitude	Example: destination.longitude=-122.141964733601	The longitude of the destination. Valid longitude values are between -180 and +180 inclusive.
destination.name	Name of the destination	Example: destination.name=Redmond, WA	The name of the destination. You do not have to encode the destination.name value.

ms-settings: parameter reference

The syntax for maps app specific parameters for the **ms-settings:** URI scheme is defined below. **maps-downloadmaps** is specified along with the **ms-settings:** URI in the form of **ms-settings:maps-downloadmaps?** to indicate the offline maps settings page.

Parameter	Definition	Example	Details
latlong	Point defining offline map region.	Example: latlong=47.6,-122.3	The geoint is specified by a comma separated latitude and longitude. Valid latitude values are between -90 and +90 inclusive. Valid longitude values are between -180 and +180 inclusive.

Launch the People app

Article • 10/20/2022

This topic describes the **ms-people**: URI scheme. Your app can use this URI scheme to launch the People app for specific actions.

ms-people: URI scheme reference

Results	URI scheme
Allows other apps to launch the People app Main page.	ms-people:
Allows other apps to launch the People app Settings page.	ms-people:settings
Allows other apps to provide a search string that will launch the People app with the result page of the search.	ms-people:search?SearchString=<contactsearchinfo>

The parameters are case sensitive.

If you do not enter the syntax correctly, or are missing the search string value, the default behavior is to return a full list of contacts without any filtering.

Results	URI scheme
Launches to an existing contact card, if the contact is found. Or, launches to a temporary contact card, if no contact is found. If no input parameter is supplied, we will launch the People App with a contact list. The parameters are case sensitive. The order of the parameters doesn't matter. If there is more than one match, we will return the first match of the contact.	<pre>ms-people:viewcontact?ContactId= <contactid>&AggregatedId= <aggid>&PhoneNumber= <phonenum>&Email= <email>&ContactName= <name>&Contact= <contactobj></pre>
Launches to a Save-contact page within the People app to save the given contact with the supplied phone number or email address. The parameters are case sensitive. The order of the parameters doesn't matter.	<pre>ms-people:savetocontact?PhoneNumber= <phonenum>&Email= <email>&ContactName= <name></pre>
Launches to the add a new contact page within the People app to save the given contact. Use LaunchUriForResultsAsync to open the save new contact page. Using LaunchUriAsync will only launch the People app Main page. The parameters are case sensitive. The order of the parameters doesn't matter. You can use any combination of supported parameters.	<pre>ms-people:savecontacttask?PhoneNumber= <phonenum>&Email= <email>&ContactName= <name></pre>

ms-people:search: parameter reference

Parameter	Description	Example
SearchString	Optional. The search string for the contact search information. The phone number or the contact name.	ms-people:search? SearchString=Smith

ms-people:viewcontact: parameter reference

Parameter	Description	Example
ContactId	Optional. Contact Id of the contact.	ms-people:viewcontact? ContactId={ContactId}
PhoneNumber	Optional. Phone number of the contact.	ms-people:viewcontact? PhoneNumber=%2014257069326
Email	Optional. Email of the contact.	ms-people:viewcontact? Email=johnsmith@contsco.com
ContactName	Optional. Name of the contact.	ms-people:viewcontact? ContactName=John%20%Smith
Contact	Optional. Contact object.	ms-people:viewcontact?Contact={Serialized Contact}

ms-people:savetocontact: parameter reference

Parameter	Description	Example
PhoneNumber	Optional. Phone number of the contact.	ms-people:savetocontact? PhoneNumber=%2014257069326

Parameter	Description	Example
Email	Optional. Email of the contact.	ms-people:savetocontact? Email=johnsmith@contsco.com
ContactName	Optional. Name of the contact.	ms-people:savetocontact? Email=johnsmith@contsco.com&ContactName=John%20%Smith

ms-people:savecontacttask: parameter reference

Parameter	Description
Company	Optional. Company name of the contact.
FirstName	Optional. First name of the contact.
HomeAddressCity	Optional. City of the home address.
HomeAddressCountry	Optional. Country of the home address.
HomeAddressState	Optional. State of the home address.
HomeAddressStreet	Optional. Street of the home address.
HomeAddressZipCode	Optional. Zip Code of the home address.
HomePhone	Optional. Home phone of the contact.
JobTitle	Optional. Job title of the contact.

Parameter	Description
LastName	Optional. Last name of the contact.
MiddleName	Optional. Middle name of the contact.
MobilePhone	Optional. Mobile phone number of the contact.
Nickname	Optional. Nickname of the contact.
Notes	Optional. Notes about the contact.
OtherEmail	Optional. Other Email of the contact.
PersonalEmail	Optional. Personal Email of the contact.
Suffix	Optional. Suffix of the contact.
Title	Optional. Title of the contact.
Website	Optional. Website of the contact.
WorkAddressCity	Optional. City of the work address.
WorkAddressCountry	Optional. Country of the work address.
WorkAddressState	Optional. State of the work address.

Parameter	Description
WorkAddressStreet	Optional. Street of work address.
WorkAddressZipCode	Optional. Zip Code of the work address.
WorkEmail	Optional. Work Email of the contact.
WorkPhone	Optional. Work phone number of the contact.

Launch screen snipping

Article • 09/12/2023

The **ms-screenclip:** and **ms-screensketch:** URI schemes allows you to initiate snipping or editing screenshots.

Open a new snip from your app

The **ms-screenclip:** URI allows your app to automatically open up and start a new snip. The resulting snip is copied to the user's clipboard, but is not automatically passed back to the opening app.

ms-screenclip: takes the following parameters:

[Expand table](#)

Parameter	Type	Required	Description
source	string	no	A freeform string to indicate the source that launched the URI.
type	string	no	A string value to indicate which special type of capture is requested. This parameter can be omitted when starting a new snip. Values supported include: snapshot, recording*
clippingMode	string	no	A string value to indicate the clipping type for the snip. Values supported include: Rectangle, Freeform, Window
delayInSeconds	int	no	An integer value, from 1 to 30. Specifies the delay, in full seconds, between the URI call and when snipping begins.
callbackformat	string	no	This parameter is unavailable.

* `type=recording` is available only on Windows 11 PCs with Snipping Tool version 11.2307 or newer, and only when the default handler for ms-screenclip is set to "Snipping Tool" instead of "Screen Clipping".

Launching the Snipping Tool or Snip & Sketch App

The **ms-screensketch:** URI allows you to programmatically launch the Snipping Tool app (on Windows 11) or Snip & Sketch app (on Windows 10), and open a specific image in that app for annotation.

ms-screensketch: takes the following parameters:

 Expand table

Parameter	Type	Required	Description
sharedAccessToken	string	no	A token identifying the file to open. Retrieved from SharedStorageAccessManager.AddFile . If this parameter is omitted, the app will be launched without a file open.
secondarySharedAccessToken	string	no	A string identifying a JSON file with metadata about the snip. The metadata may include a clipPoints field with an array of x,y coordinates, and/or a userActivity .
source	string	no	A freeform string to indicate the source that launched the URI.
isTemporary	bool	no	If set to True, Snipping Tool will try to delete the file after opening it.

The following example calls the [LaunchUriAsync](#) method to send an image to Snipping Tool from the user's app.

C#

```
bool result = await Windows.System.Launcher.LaunchUriAsync(new Uri("ms-screensketch:edit?source=MyApp&isTemporary=false&sharedAccessToken=2C37ADDA-B054-40B5-8B38-11CED1E1A2D"));
```

The following example illustrates what a file specified by the **secondarySharedAccessToken** parameter of **ms-screensketch** might contain:

JSON

```
{
  "clipPoints": [
    {
      "x": 0,
      "y": 0
    },
    {
      "x": 2080,
      "y": 0
    }
  ]
}
```

```

{
  "x": 2080,
  "y": 780
},
{
  "x": 0,
  "y": 780
}
],
"userActivity": "{\\"$schema\\":\\"http://activity.windows.com/user-activity.json\\",\\"UserActivity\\":\\"type\\",\\"1.0\\":\\"version\\",\\"cross-platform-identifiers\\":
[{\\"platform\\":\\"windows_universal\\",\\"application\\":\\"Microsoft.MicrosoftEdge_8wekyb3d8bbwe!MicrosoftEdge\\"},
{\\"platform\\":\\"host\\",\\"application\\":\\"edge.activity.windows.com\\"}],\\"activationUrl\\":\\"microsoft-
edge:https://support.microsoft.com/help/13776/windows-use-snipping-tool-to-capture-
screenshots\\",\\"contentUrl\\":\\"https://support.microsoft.com/help/13776/windows-use-snipping-tool-to-capture-screenshots\\",\\"visualElements\\":
{\\"attribution\\":{\\"iconUrl\\":\\"https://www.microsoft.com/favicon.ico?v2\\",\\"alternateText\\":\\"microsoft.com\\"},\\"description\\":\\"https://support.
microsoft.com/help/13776/windows-use-snipping-tool-to-capture-screenshots\\",\\"backgroundColor\\":\\"#FF0078D7\\",\\"displayText\\":\\"Use
snipping tool to capture screenshots - Windows Help\\",\\"content\\":
{\\"$schema\\":\\"http://adaptivecards.io/schemas/adaptive-
card.json\\",\\"type\\":\\"AdaptiveCard\\",\\"version\\":\\"1.0\\",\\"body\\":
[{\\"type\\":\\"Container\\",\\"items\\":[{\\"type\\":\\"TextBlock\\",\\"text\\":\\"Use
snipping tool to capture screenshots - Windows
Help\\",\\"weight\\":\\"bolder\\",\\"size\\":\\"large\\",\\"wrap\\":true,\\"maxLines\\":3
}],
{\\"type\\":\\"TextBlock\\",\\"text\\":\\"https://support.microsoft.com/help/13776/
windows-use-snipping-tool-to-capture-
screenshots\\",\\"size\\":\\"normal\\",\\"wrap\\":true,\\"maxLines\\":3}}]}\\",\\"isRoamable\\":true,\\"appActivityId\\":\\"https://support.microsoft.com/help/13776/wi
ndows-use-snipping-tool-to-capture-screenshots\\"}"
}

```

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Enable apps for websites using app URI handlers

Article • 01/03/2023

Apps for Websites associates your app with a website so that when someone opens a link to your website, your app is launched instead of opening the browser. If your app is not installed, your website opens in the browser as usual. Users can trust this experience because only verified content owners can register for a link. Users will be able to check all of their registered web-to-app links by going to Settings > Apps > Apps for websites.

To enable web-to-app linking you will need to:

- Identify the URIs your app will handle in the manifest file
- A JSON file that defines the association between your app and your website. with the app Package Family Name at the same host root as the app manifest declaration.
- Handle the activation in the app.

ⓘ Note

Starting with the Windows 10 Creators update, supported links clicked in Microsoft Edge Legacy will launch the corresponding app. Supported links clicked in other browsers (for example, Microsoft Edge Chromium, Internet Explorer, etc.), will keep you in the browsing experience.

Register to handle http and https links in the app manifest

Your app needs to identify the URIs for the websites it will handle. To do so, add the **Windows.appUriHandler** extension registration to your app's manifest file **Package.appxmanifest**.

For example, if your website's address is "msn.com" you would make the following entry in your app's manifest:

XML

```
<Applications>
  <Application ... >
  ...
```